

The vector and Banach lattices library

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Chapter 1

The Basics

The results the reader may find missing in this chapter are likely already part of `mathlib`.

1.1 Vector lattices and the Archimedean property

This section sets up the basic order-theoretic algebra underpinning the rest of the library. We work first in the generality of a lattice-ordered abelian group, establishing the fundamental identities for the positive part x^+ , the negative part x^- , and the absolute value $|x|$, together with the uniqueness of the positive part and the interaction of the lattice operations with translation and with arbitrary suprema and infima. We then specialise to *vector lattices*, i.e. real vector spaces with a compatible lattice structure, and record how non-negative scalars distribute over the lattice operations, culminating in the identity $|\lambda x| = |\lambda| |x|$. The section closes with the Archimedean property, which plays a foundational role in later chapters: we give two equivalent formulations and use them to show that an Archimedean vector lattice of rank at least two necessarily has vectors which are neither positive nor negative.

1.1.1 Lattice-ordered abelian groups

Throughout this subsection X denotes an additive abelian group equipped with a lattice order compatible with addition (an ℓ -group). We write $x^+ := x \sqcup 0$, $x^- := (-x) \sqcup 0$ and $|x| := x \sqcup (-x)$.

Lemma 1.1.1. *Let X be a lattice-ordered abelian group and let $x \in X$. Then $|x| = 0$ if and only if $x = 0$.*

Lemma 1.1.2. *Let X be a lattice-ordered abelian group, let $x \in X$, and suppose $u, v \in X$ satisfy $x = u - v$ and $u \sqcap v = 0$. Then $u = x^+$.*

Lemma 1.1.3. *For all x, y in a lattice-ordered abelian group, $x \sqcup y = x + (y - x)^+$.*

Lemma 1.1.4. *For all x, y in a lattice-ordered abelian group, $x \sqcap y = x - (x - y)^+$.*

Lemma 1.1.5. *For all x, y in a lattice-ordered abelian group, $x - x \sqcap y = (x - y)^+$.*

Lemma 1.1.6. *For all x, y in a lattice-ordered abelian group, $(x + y)^+ \leq x^+ + y^+$.*

Lemma 1.1.7. *Let X be a lattice-ordered abelian group and let $x, a, b \in X$ with $0 \leq x$, $0 \leq a$, and $0 \leq b$. Then $x \sqcap (a + b) \leq x \sqcap a + x \sqcap b$.*

Lemma 1.1.8. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x + a = \sup\{x + z : z \in A\}$.*

Lemma 1.1.9. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x + a = \inf\{x + z : z \in A\}$.*

Lemma 1.1.10. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x \sqcap a = \sup\{x \sqcap z : z \in A\}$.*

Lemma 1.1.11. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x \sqcup a = \inf\{x \sqcup z : z \in A\}$.*

Lemma 1.1.12. *Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a + b = \inf\{p + q : p \in A, q \in B\}$.*

Lemma 1.1.13. *Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \sqcup b = \inf\{p \sqcup q : p \in A, q \in B\}$.*

Lemma 1.1.14. *Let X be a lattice, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \sqcap b = \inf(A \cup B)$.*

1.1.2 Vector lattices

A *vector lattice* is a lattice-ordered abelian group equipped with a real vector space structure whose scalar multiplication by non-negative reals is monotone. Throughout this subsection X denotes a vector lattice.

Definition 1.1.15. Let X be an additive abelian group equipped with a lattice structure compatible with addition. A *vector lattice* structure on X consists of a real vector space structure on X for which scalar multiplication by non-negative reals is monotone: for all $\lambda \geq 0$ and $x \leq y$ in X , $\lambda x \leq \lambda y$.

Lemma 1.1.16. *Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \sqcup y) = (\lambda x) \sqcup (\lambda y)$.*

Lemma 1.1.17. *Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \sup A$, then $\lambda a = \sup\{\lambda z : z \in A\}$.*

Lemma 1.1.18. *Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \inf A$, then $\lambda a = \inf\{\lambda z : z \in A\}$.*

Lemma 1.1.19. *Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \sqcap y) = (\lambda x) \sqcap (\lambda y)$.*

Lemma 1.1.20. *Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \sqcup \beta) \cdot x = (\alpha x) \sqcup (\beta x)$.*

Lemma 1.1.21. *Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \sqcap \beta) \cdot x = (\alpha x) \sqcap (\beta x)$.*

Theorem 1.1.22. *Let X be a vector lattice, let $x \in X$ and $\alpha \in \mathbb{R}$. Then $|\alpha x| = |\alpha| \cdot |x|$.*

Lemma 1.1.23. *Let X be a vector lattice, let $x, y \in X$ with $x \sqcap y = 0$ and let $\lambda \in \mathbb{R}$ with $0 \leq \lambda$. Then $(\lambda x) \sqcap y = 0$.*

1.1.3 The Archimedean property

Definition 1.1.24. A lattice-ordered abelian group X is called *Archimedean* when, for every $x, y \in X$ with $0 \leq x$ and $n \cdot x \leq y$ for all $n \in \mathbb{N}$, one has $x = 0$.

Lemma 1.1.25. Let X be a lattice-ordered abelian group. Then X is Archimedean if and only if for all $x, y \in X$ the condition $n \cdot x \leq y$ for all $n \in \mathbb{N}$ implies $x \leq 0$.

Lemma 1.1.26. A vector lattice X is Archimedean if and only if for every $u \in X$ with $0 \leq u$, $\inf\{n^{-1} \cdot u : n \in \mathbb{N}, n > 0\} = 0$.

Lemma 1.1.27. Let X be an Archimedean lattice-ordered abelian group and let $x, y \in X$. If $n \cdot |x| \leq y$ for all $n \in \mathbb{N}$, then $x = 0$.

Theorem 1.1.28. Let X be an Archimedean vector lattice with real-vector-space rank strictly greater than 1. Then there exists $x \in X$ which is neither non-negative nor non-positive.

1.2 Riesz decomposition

This section collects the Riesz decomposition theorem and its standard refinements in a lattice-ordered abelian group X . The binary form splits a positive element dominated by a sum of two positive elements into a corresponding sum. Iterating and combining the argument yields a matrix form for two partitions of the same positive sum, a version with a finite sum on the left hand side, a decomposition adapted to the absolute value, and an identification of sum intervals $[0, x] + [0, y] = [0, x + y]$ for $x, y \geq 0$.

Theorem 1.2.1. Let X be a lattice-ordered abelian group and let $x, y, z \in X$ with $0 \leq x$, $0 \leq y$, $0 \leq z$ and $x \leq y + z$. Then there exist $x_1, x_2 \in X$ with $0 \leq x_1 \leq y$, $0 \leq x_2 \leq z$ and $x = x_1 + x_2$.

Theorem 1.2.2. Let X be a lattice-ordered abelian group, let $m, n \in \mathbb{N}$, let $u : \{0, \dots, m-1\} \rightarrow X$ and $z : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq u_i$ for all i and $0 \leq z_j$ for all j , and $\sum_i u_i = \sum_j z_j$. Then there exists a family $x : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \rightarrow X$ with $0 \leq x_{ij}$ for all i, j such that $u_i = \sum_j x_{ij}$ for every i and $z_j = \sum_i x_{ij}$ for every j .

Theorem 1.2.3. Let X be a lattice-ordered abelian group, let $k \in \mathbb{N}$, let $u : \{0, \dots, k-1\} \rightarrow X$ and $x, y \in X$ with $0 \leq u_i$ for all i , $0 \leq x$, $0 \leq y$ and $\sum_i u_i \leq x + y$. Then there exist $v, w : \{0, \dots, k-1\} \rightarrow X$ with $0 \leq v_i$ and $0 \leq w_i$ for all i , $\sum_i v_i \leq x$, $\sum_i w_i \leq y$, and $u_i = v_i + w_i$ for every i .

Theorem 1.2.4. Let X be a lattice-ordered abelian group, let $x, u, v \in X$ with $0 \leq u$, $0 \leq v$ and $|x| = u + v$. Then there exist $y, z \in X$ with $x = y + z$, $|y| = u$ and $|z| = v$.

Theorem 1.2.5. Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$, let $x \in X$ and let $v : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq v_i$ for every i and $|x| \leq \sum_i v_i$. Then there exists $y : \{0, \dots, n-1\} \rightarrow X$ with $x = \sum_i y_i$ and $|y_i| \leq |x| \sqcap v_i$ for every i .

Theorem 1.2.6. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. Then $[0, x] + [0, y] = [0, x + y]$ as subsets of X .

1.3 Normed and Banach lattices

A *normed vector lattice* is a real vector lattice equipped with a norm satisfying the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$. This compatibility between the norm and the lattice operations has a number of immediate consequences: the lattice operations $\sqcup$, $\sqcap$ and $|\cdot|$ are norm-continuous, the space is automatically Archimedean, the positive cone is norm-closed, order intervals are norm-closed and norm-bounded, and monotone norm-convergent sequences converge to their supremum. A *Banach lattice* is a normed vector lattice whose norm is complete.

Definition 1.3.1. Let X be a normed abelian group equipped with a compatible lattice structure. A *normed vector lattice* structure on X consists of a vector lattice structure together with the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$ for all $x, y \in X$, and the identity $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$ and $x \in X$.

Lemma 1.3.2. Every normed vector lattice is a normed space over $\mathbb{R}$.

Theorem 1.3.3. Let X be a normed vector lattice. The map $(x, y) \mapsto x \sqcup y$ from $X \times X$ to X is continuous.

Theorem 1.3.4. Let X be a normed vector lattice. The map $(x, y) \mapsto x \sqcap y$ from $X \times X$ to X is continuous.

Theorem 1.3.5. Let X be a normed vector lattice. The map $x \mapsto |x|$ from X to X is Lipschitz with constant 1.

Theorem 1.3.6. Every normed vector lattice is Archimedean.

Lemma 1.3.7. Let X be a normed vector lattice. The order relation on X is closed in the norm topology.

Theorem 1.3.8. Let X be a normed vector lattice. The positive cone $\{x \in X : 0 \leq x\}$ is norm-closed.

Theorem 1.3.9. Let X be a normed vector lattice, let $u, v : \mathbb{N} \rightarrow X$ and $a, b \in X$. If $u_n \rightarrow a$ and $v_n \rightarrow b$ in norm and $u_n \leq v_n$ for every n , then $a \leq b$.

Theorem 1.3.10. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be monotone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the least upper bound of the range of u .

Theorem 1.3.11. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be antitone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the greatest lower bound of the range of u .

Lemma 1.3.12. Let X be a normed vector lattice and let $a, b \in X$. The order interval $[a, b]$ is norm-closed.

Lemma 1.3.13. Let X be a normed vector lattice, let $a, b \in X$ and let $x \in [a, b]$. Then $\|x\| \leq \|a\| \sqcup \|b\|$.

Theorem 1.3.14. Let X be a normed vector lattice and let $s \subseteq X$ be both bounded below and bounded above in the order. Then s is norm-bounded.

Definition 1.3.15. A *Banach lattice* is a normed vector lattice whose underlying normed space is complete.